

LUCAS LÉVY

lucas.levy@inria.fr — <https://lucas-levy.github.io>

EDUCATION

- **École Normale Supérieure Paris-Saclay** 2025 – 2026
Master Mathématiques, Vision, Apprentissage (MVA)
 - Relevant coursework includes *Convex Optimization*, *Stopping Times and Online Algorithms*, *Computational Statistics*, *Representation Learning*, *Kernel methods*, and *Learning for Time Series*
- **École Polytechnique** 2022 – 2026
Cycle ingénieur polytechnicien (X2022)
 - French top-ranking engineering school, delivering a Master-level French Engineer’s degree
 - Specialization in Applied Mathematics
 - GPA: 3.90/4.00
 - Relevant coursework includes *Foundations of ML*, *Statistics*, *Deep Learning*, *Optimization*, and *Algorithmics*
- **Université Paris-Nanterre** 2023 – 2024
Bachelor’s degree, History
 - Bachelor’s degree obtained in parallel with my curriculum at École Polytechnique
- **Lycée Louis-le-Grand** 2020 – 2022
*CPGE MPSI-MP**
 - Intensive scientific preparation for Polytechnique entrance examination

EXPERIENCE

- **Inria Paris** 04/2026 – Present
Research Intern Paris, France
 - Research in the SIERRA team, working under the supervision of Michael I. Jordan and Francis Bach.
- **University of Oxford** 03 – 08/2025
Visiting Research Student Oxford, United Kingdom
 - Department of Statistics – Supervisors: Dr. Arya Akhavan and Prof. Patrick Rebeschini
 - Conducting research on online linear optimization under bandit feedback, developing a novel algorithm combining self-concordant barriers properties and perturbation-based regularization
 - Performing rigorous theoretical analysis using tools from optimization, statistics, and stochastic process theory
 - Results published to Conference on Learning Theory (COLT 2026)
 - Research distinguished with *congratulations from the jury* of Polytechnique (< 5% of research internships)
- **Wandercraft** 06 – 08/2024
R&D Machine Learning Intern Paris, France
 - Applying machine learning and time-series analysis for online fault detection in a self-stabilized exoskeleton
 - Mastering tools and techniques in operational ML such as python libraries PyTorch and Scikit-learn

PUBLICATIONS

- **Lévy, L.**, Valeau, J.-L., Akhavan, A., and Rebeschini, P. (2025). Self-concordant perturbations for linear bandits. *Accepted at COLT 2026*.

PROJECT WORK

- **Implementation Project**

01 – 03/2025

École Polytechnique

- Project in Deep Learning course on *Attention Rollout and Flow for Explainability in Vision Transformers*
- Exploring methods for quantifying the flow of information through self-attention in neural networks
- Implementing and simulating these methods using PyTorch

- **Team Research Project**

09 – 12/2024

EA Recherche, École Polytechnique

- *Online Optimization of a Battery Profitability with Reserve Commitment*, under the guidance of Prof. Frédéric Meunier
- Developing and analyzing online optimization algorithms for battery energy storage in electricity markets
- Applying competitive analysis and dynamic programming to ensure robust performance under uncertainty

REFEREES

- Prof. Patrick Rebeschini, Department of Statistics, University of Oxford, patrick.rebeschini@stats.ox.ac.uk
- Dr. Arya Akhavan, Department of Statistics, University of Oxford, arya.akhavan@stats.ox.ac.uk

ADDITIONAL SKILLS

- **Languages:** French (Native), English (Fluent, *Linguaskill Certification C1+ Level*)
- **Programming skills:**
 - Python, C++, Java
 - Python libraries for ML and data analysis: PyTorch, Scikit-learn, pandas, numpy